CloudGuard IoT — Problem Statement

Project: Unified Autonomous Cloud & IoT Security Platform

Document type: System Analysis & Design

Status: Initial project baseline

Version: 1.0

1. Background

Organizations increasingly use cloud infrastructure and Internet of Things (IoT) devices together. Cloud resources host critical APIs, storage, and databases, while IoT devices send telemetry via protocols like MQTT. Monitoring these environments separately makes it difficult for operators to determine if isolated events are part of a coordinated cyber attack.

2. Problem Statement

The project addresses the challenge of achieving unified visibility and coordinated incident handling across selected AWS cloud resources and virtual or registered IoT devices. The prototype collects security events, evaluates suspicious behavior with explainable rules, correlates related findings, and triggers predefined responses while maintaining full auditability.

3. Main Challenges

- **Fragmented visibility:** Security data split across disconnected monitoring consoles.
- **Difficult event correlation:** Hard to link IoT anomalies to cloud infrastructure events.
- **Inconsistent prioritization:** Lack of automated, rule-based risk scoring.
- **Manual containment:** Slow manual response during active device or resource compromises.
- **Investigation gaps:** Disjointed logs and missing timeline synchronization during post-incident analysis.

4. Proposed Solution

Monitor → Detect → Assess Risk → Correlate → Respond → Preserve Evidence → Recover

CloudGuard IoT delivers an automated, rule-based security engine that bridges AWS IoT Core telemetry with CloudTrail/Security findings into a unified operational dashboard.

5. Intended Users

The primary intended users are project evaluators, security operators, and administrators who need to review monitored resources, investigate incidents, and inspect response outcomes. The prototype also provides a read-only view for demonstration purposes.

6. Expected Outcome

A demonstrable prototype capable of executing controlled test scenarios showing:

- Ingestion of cloud and IoT telemetry.
- Rule-based identification of suspicious behaviors.
- Calculation and transparent display of risk scores.
- Automated grouping of related findings into unified incidents.
- Execution and recording of authorized response actions.
- Interactive timeline investigation through a single dashboard.

7. Success Conditions

The project will be considered successful when the team can:

- Demonstrate end-to-end detection and response in an authorized test environment.
- Produce repeatable test results across multiple consecutive test runs.
- Document known operational limitations and system constraints honestly.
- Explain the underlying system architecture and security controls to evaluators.